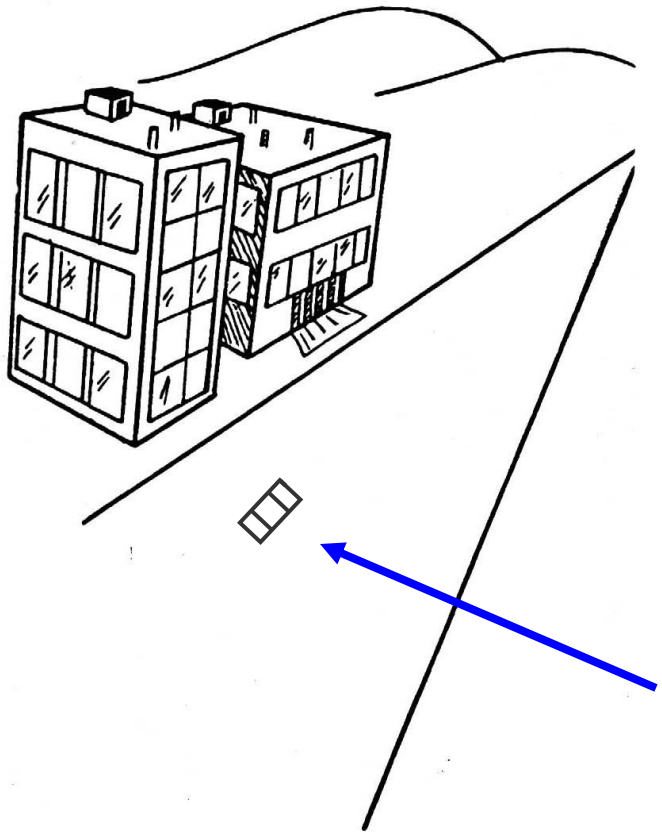
Image Segmentation

How many zebras?

From Sandlot Science

Why context is important?

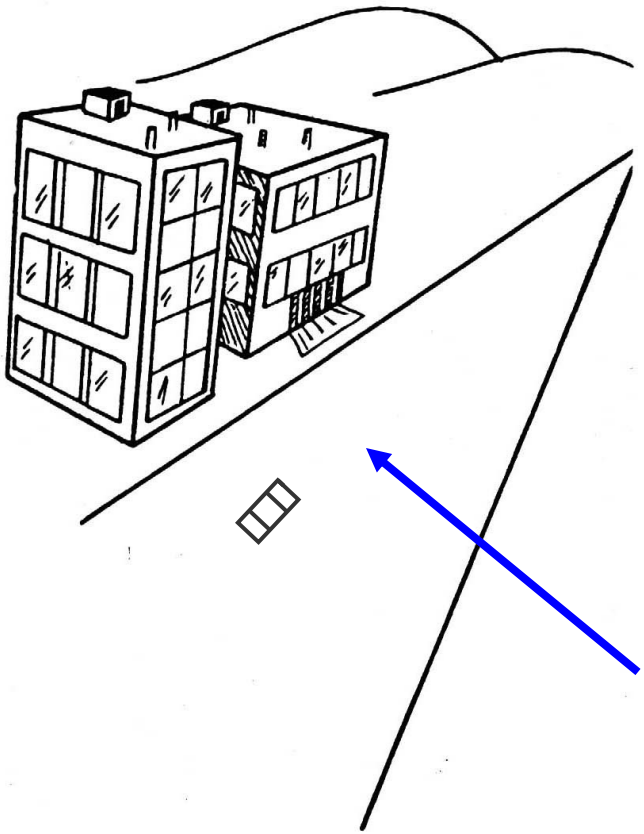


What is this?

slide by Takeo Kanade

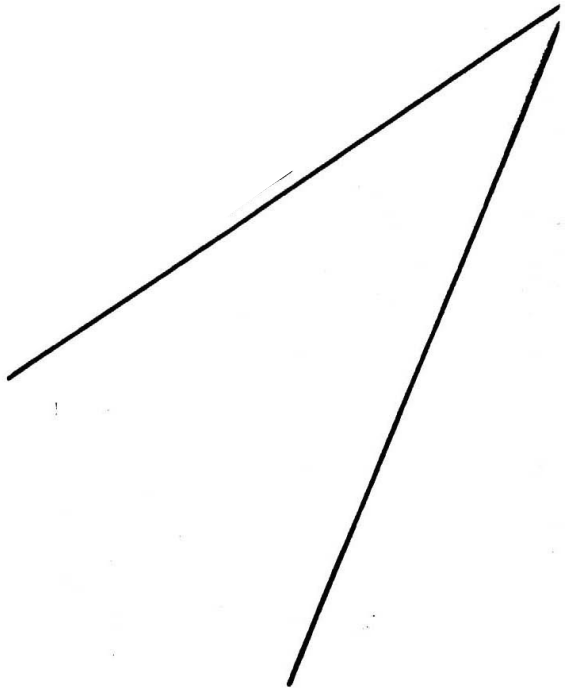
Why is this a car?

…because it’s on the road!



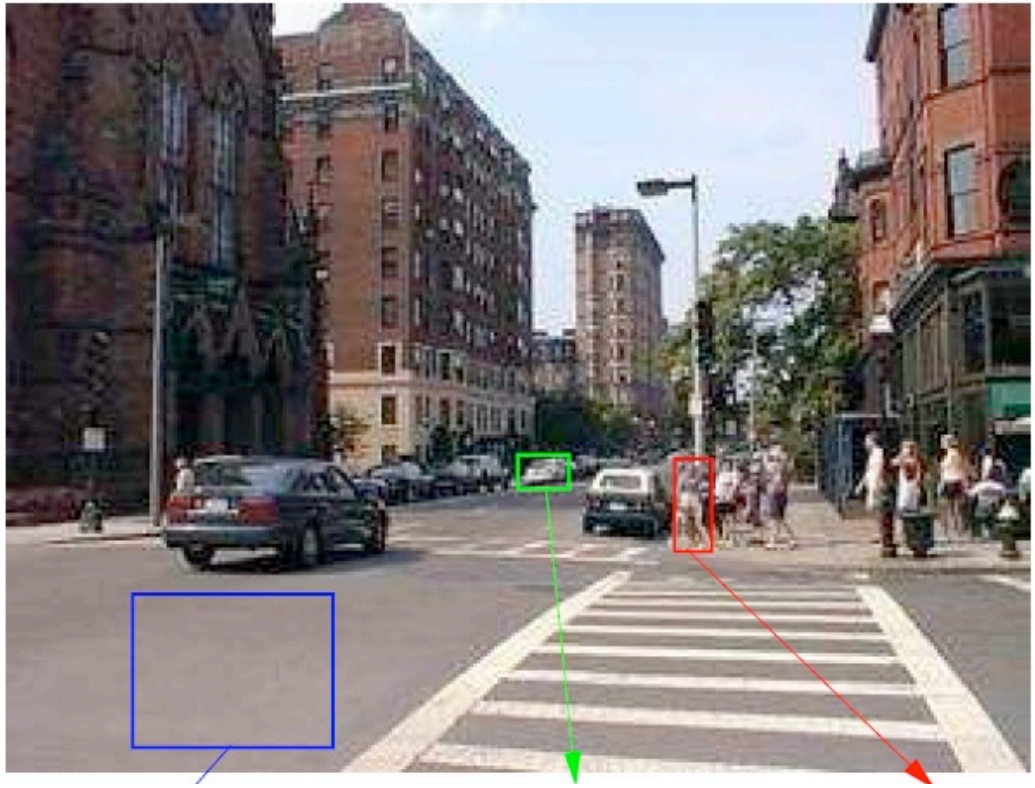
Why is this road?

Why is this a road?

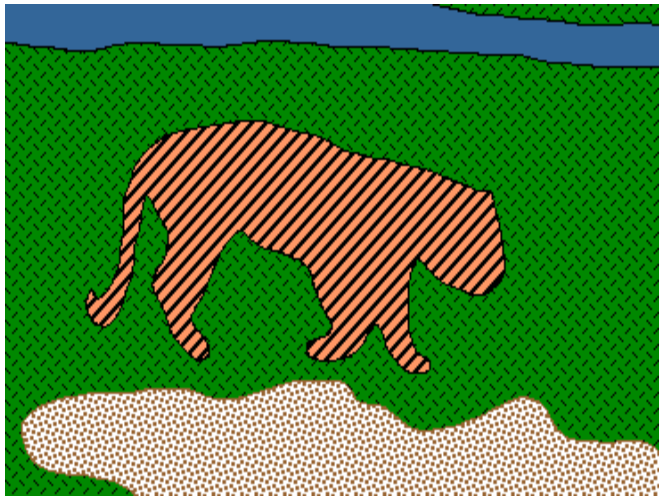


Context is very important!

Same problem in real scenes



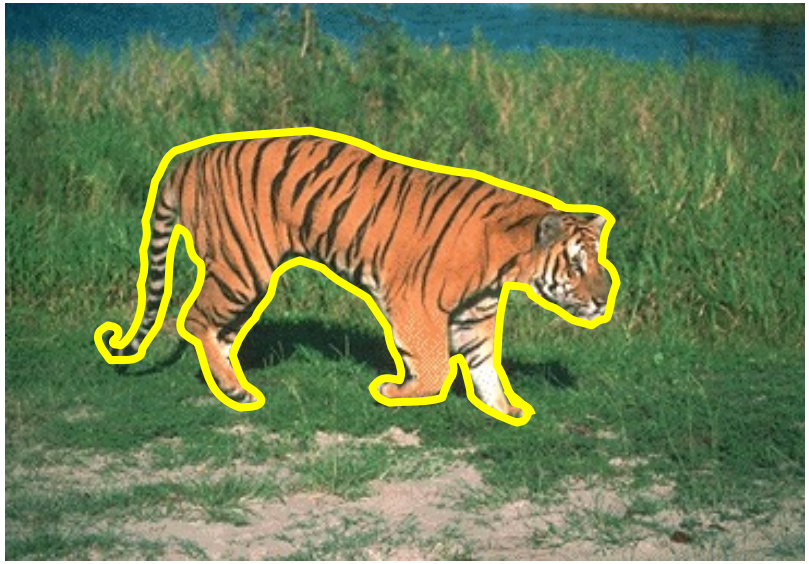
From images to objects



What defines an object?

- Subjective problem, but has been well-studied

Extracting objects



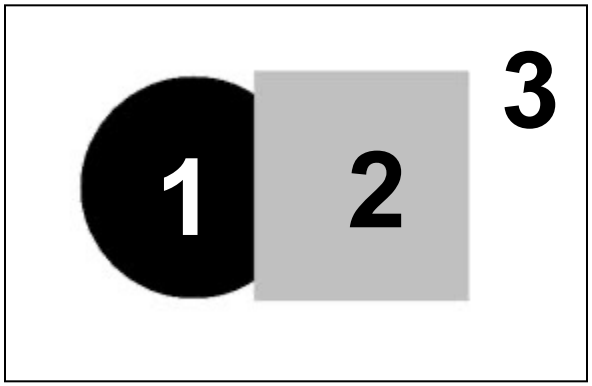
How could we do this automatically (or at least semi-automatically)?

Semi-automatic binary segmentation

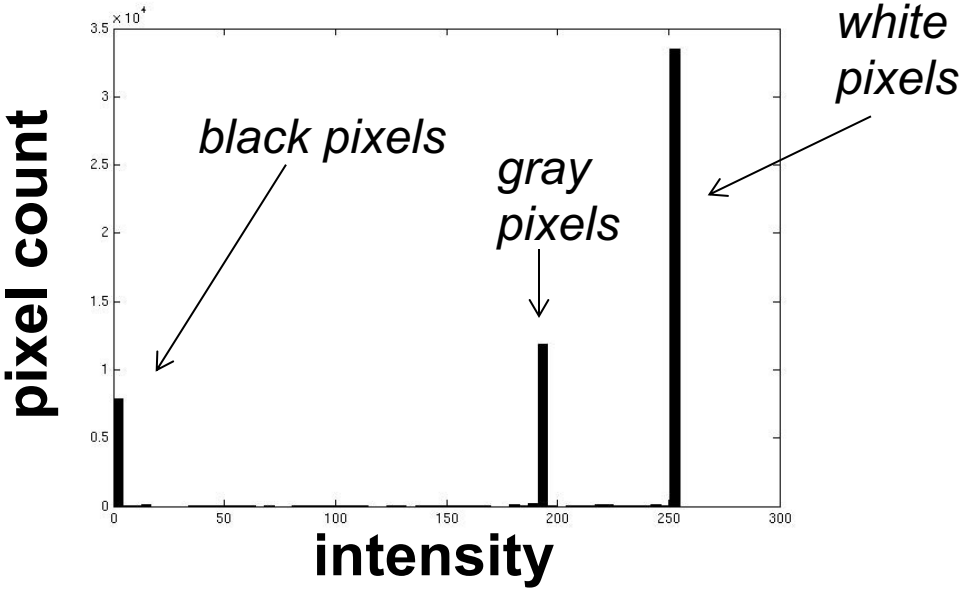
Simplifying the user interaction

Grabcut [Rother et al., SIGGRAPH 2004]

Auto segmentation: toy example

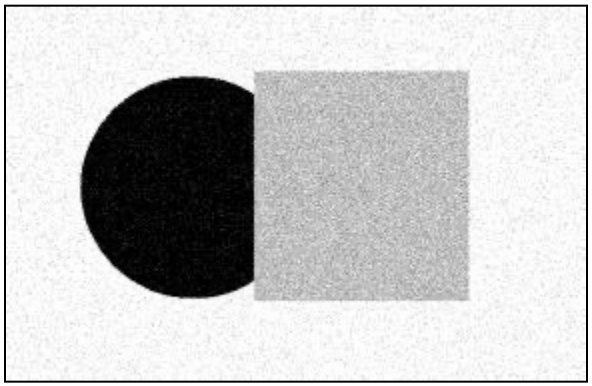


input image

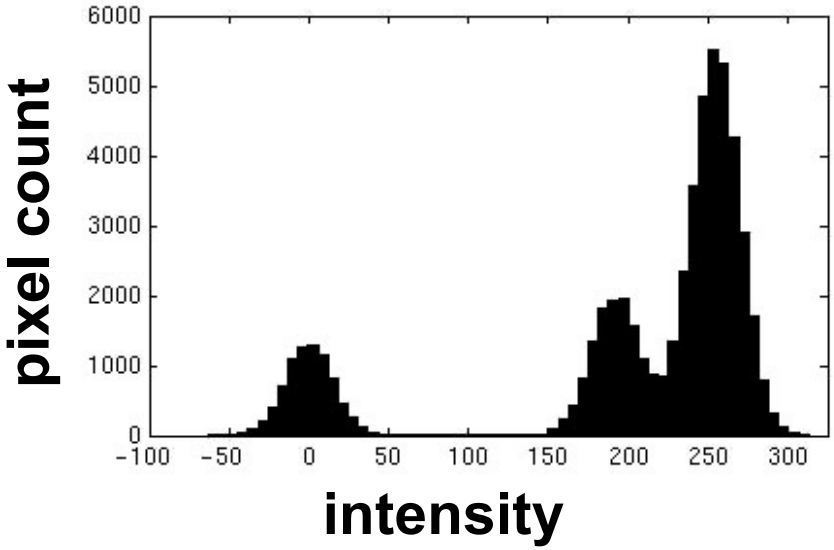


- These intensities define the three groups.
- We could label every pixel in the image according to which of these primary intensities it is.
  - i.e., *segment* the image based on the intensity feature.
- But … image isn't quite so simple …

Source: K. Grauman



input image

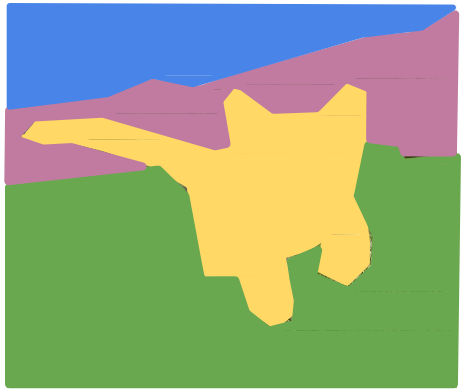


- Now how to determine the three main intensities that define our groups?
- We need to ***cluster.***

Source: K. Grauman

Deep Learning

Semantic Segmentation



GRASS, CAT, TREE, SKY

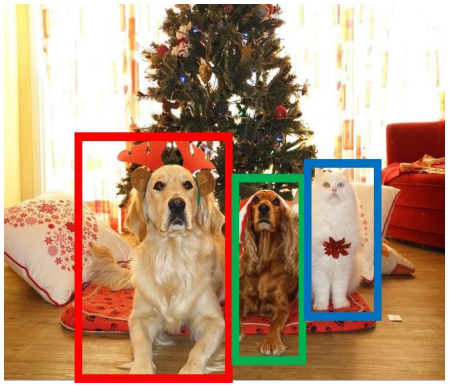
Pixel-level

Classification + Localization

CAT

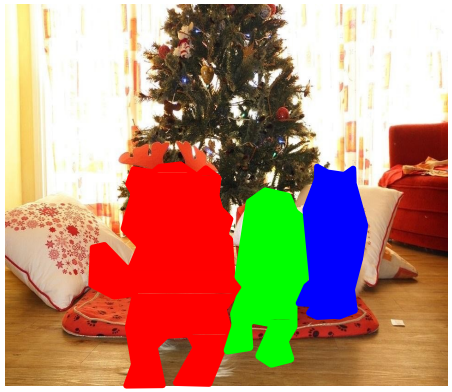
Single Object

Object Detection



DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

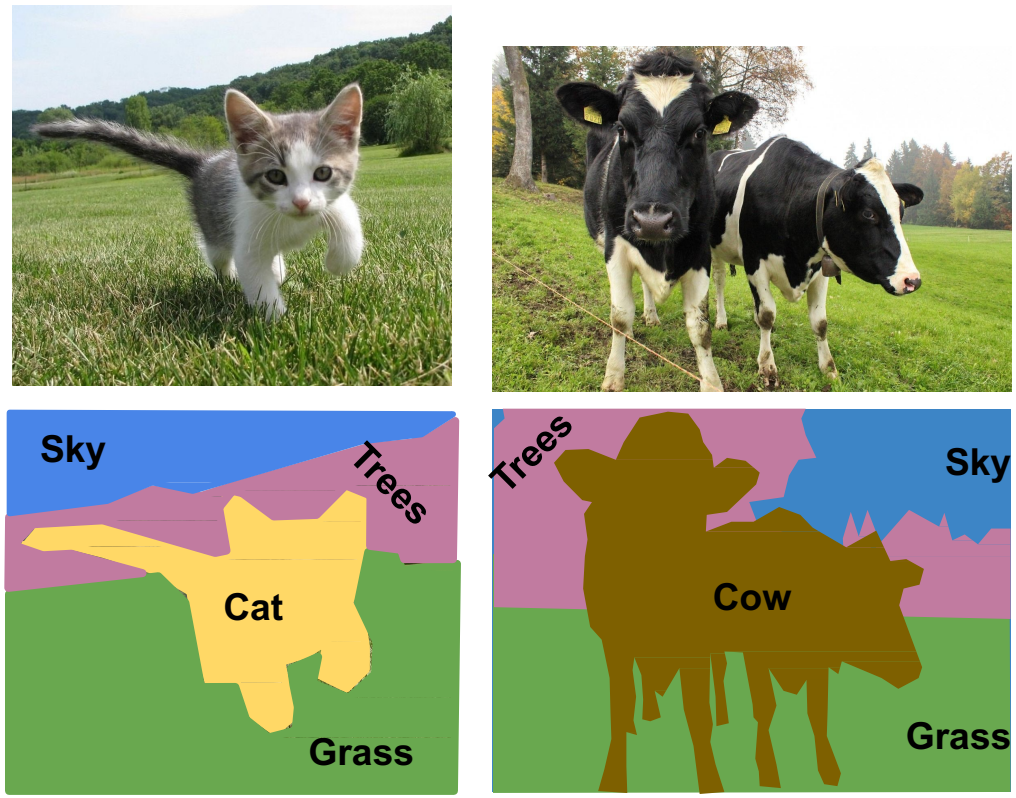
Multiple Object

Segmentation+Classification

Slide by: Justin Johnson

Semantic Segmentation

Label each pixel in the image with a category label



Don’t differentiate instances, only care about pixels

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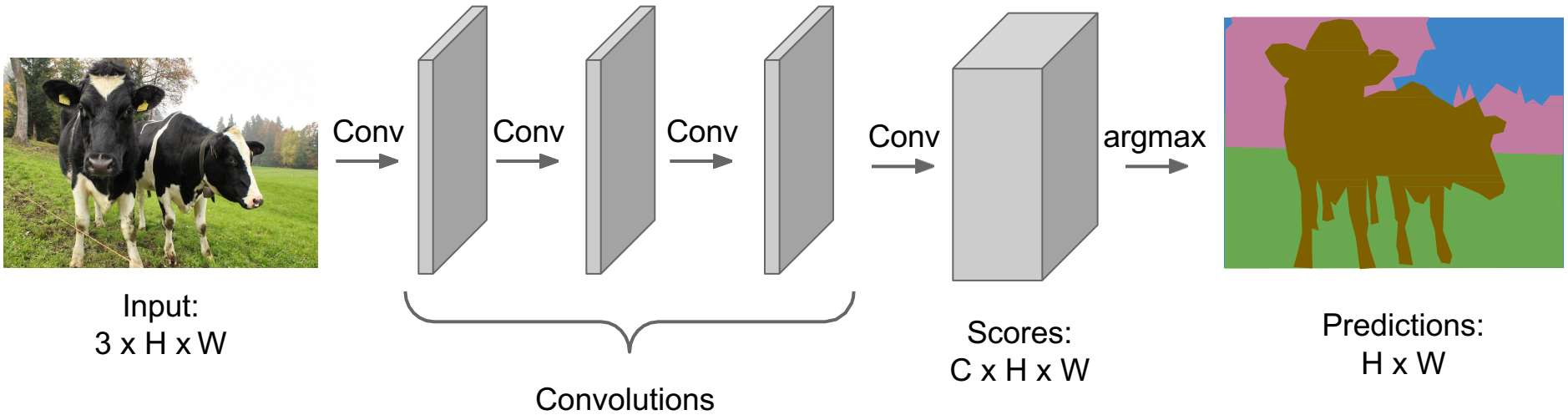
Semantic Segmentation Idea: Fully Convolutional

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



Each channel is a class
C channels->C classes

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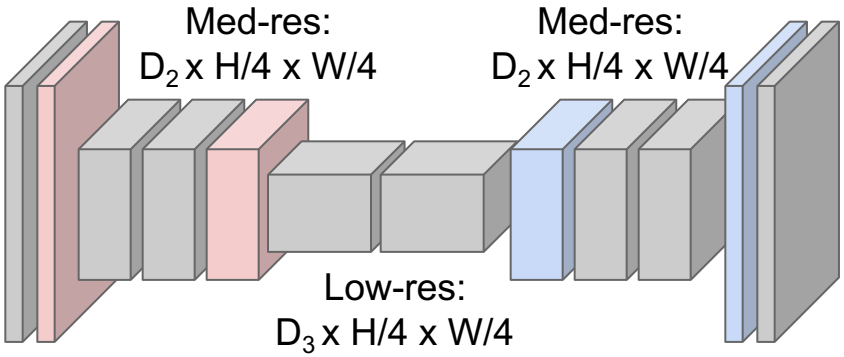
Semantic Segmentation Idea: Fully Convolutional

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Input:
3 x H x W



High-res:
D_1 x H/2 x W/2

High-res:
D_1 x H/2 x W/2

Predictions:
H x W

Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

Slide by: Justin Johnson

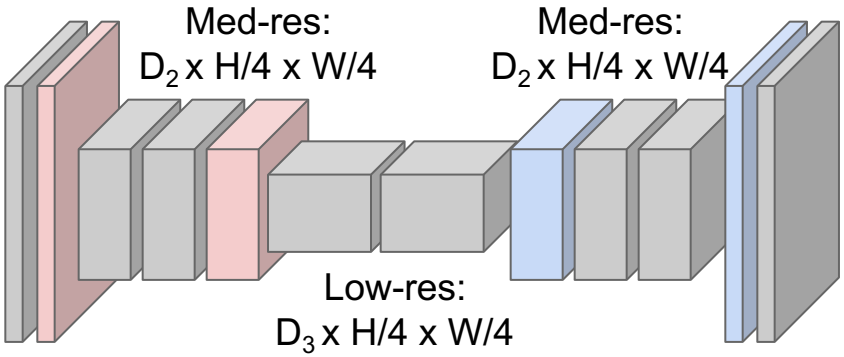
Semantic Segmentation Idea: Fully Convolutional

Downsampling: Pooling, strided convolution

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Upsampling: ???

Input:
3 x H x W



High-res:
D_1 x H/2 x W/2

High-res:
D_1 x H/2 x W/2

Predictions:
H x W

Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015

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In-Network upsampling: “Unpooling”

Nearest Neighbor

1	2
3	4

Input: 2 x 2

1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Output: 4 x 4

“Bed of Nails”

1	2
3	4

Input: 2 x 2

1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

Output: 4 x 4

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In-Network upsampling: “Max Unpooling”

Max Pooling
Remember which element was max!

1	2	6	3
3	5	2	1
1	2	2	1
7	3	4	8

Input: 4 x 4

5	6
7	8

Output: 2 x 2

…

Rest of the network

Max Unpooling
Use positions from pooling layer

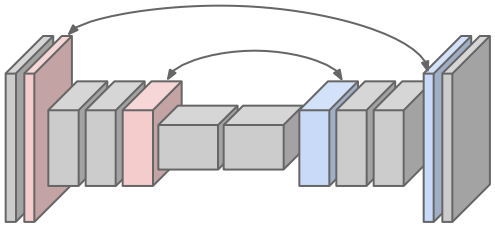
1	2
3	4

Input: 2 x 2

0	0	2	0
0	1	0	0
0	0	0	0
3	0	0	4

Output: 4 x 4

Corresponding pairs of downsampling and upsampling layers



Slide by: Justin Johnson

Learnable Upsampling

3 x 3 **transpose** convolution, stride 2 pad 1

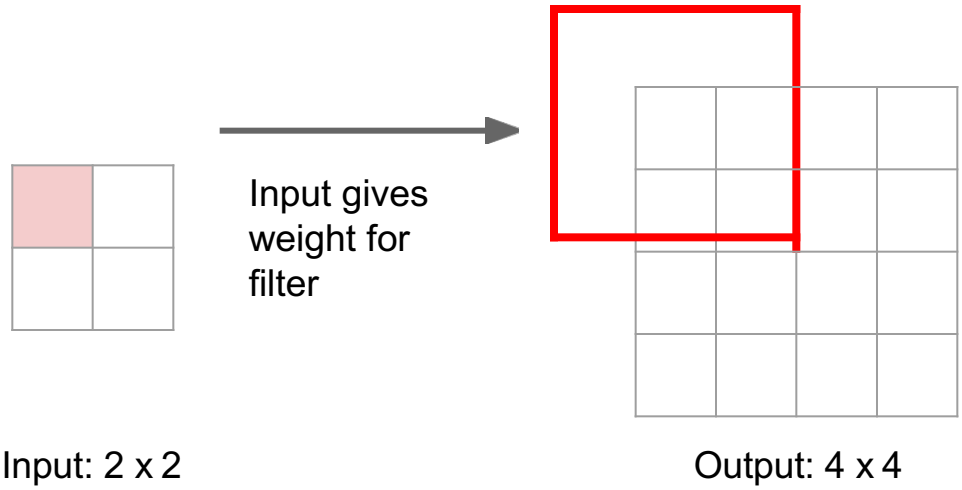
Input: 2 x 2

Output: 4 x 4

Slide by: Justin Johnson

Learnable Upsampling

3 x 3 **transpose** convolution, stride 2 pad 1



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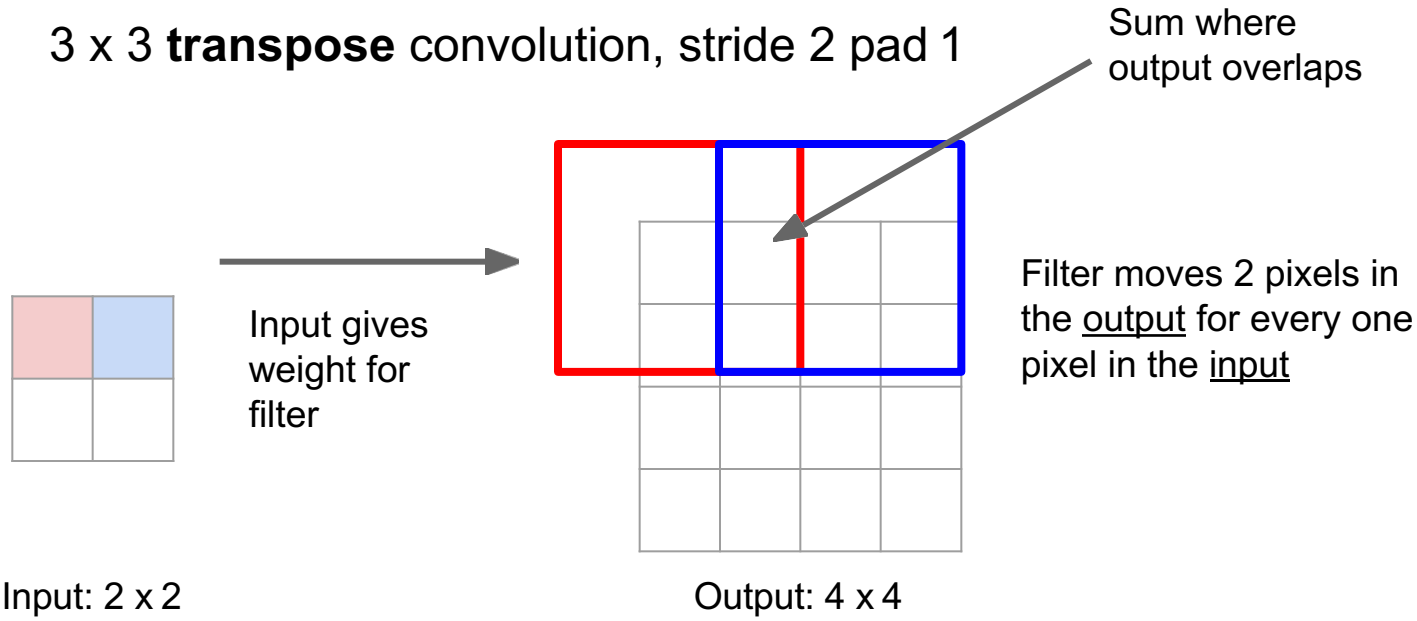
Learnable Upsampling

3 x 3 **transpose** convolution, stride 2 pad 1



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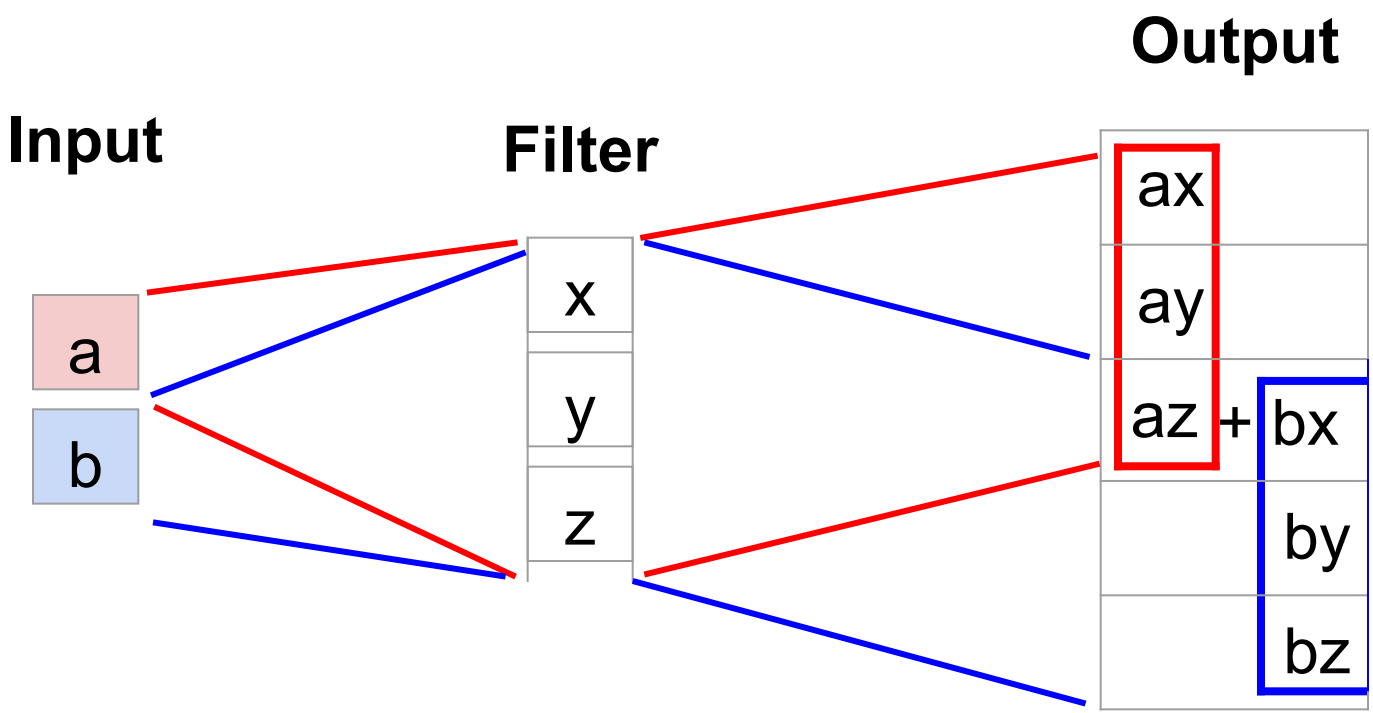
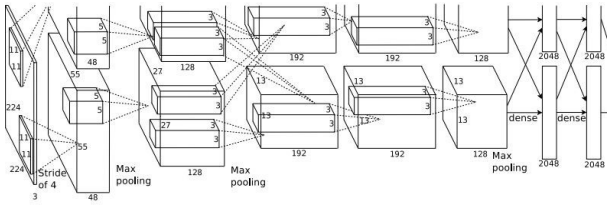
Transpose Convolution: 1D Example



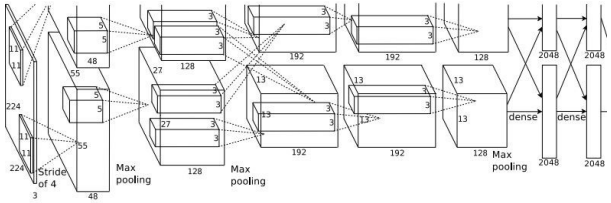
Output contains copies of the filter weighted by the input, summing at where at overlaps in the output

Adapted from Justin Johnson

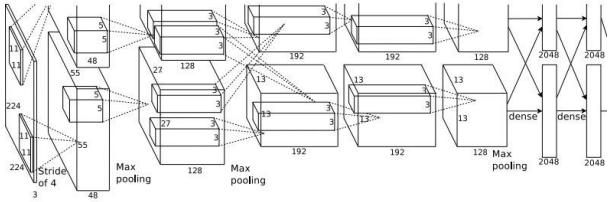
Object Detection as Regression?



CAT: (x, y, w, h)



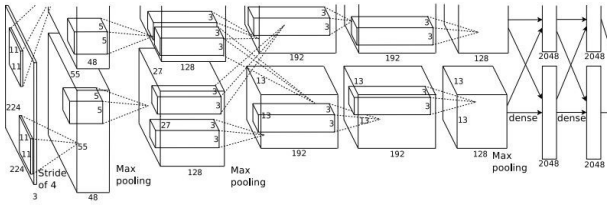
DOG: (x, y, w, h)
DOG: (x, y, w, h)
CAT: (x, y, w, h)



DUCK: (x, y, w, h)
DUCK: (x, y, w, h)
….

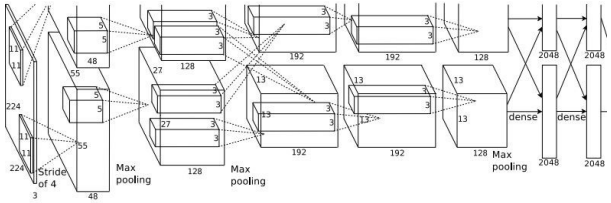
Slide by: Justin Johnson

Object Detection as Regression?



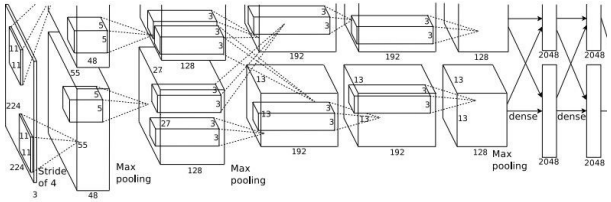
CAT: (x, y, w, h)

4 numbers



DOG: (x, y, w, h)
DOG: (x, y, w, h)
CAT: (x, y, w, h)

16 numbers



DUCK: (x, y, w, h)
DUCK: (x, y, w, h)
….

Many numbers!

Each image needs a different number of outputs!

Slide by: Justin Johnson